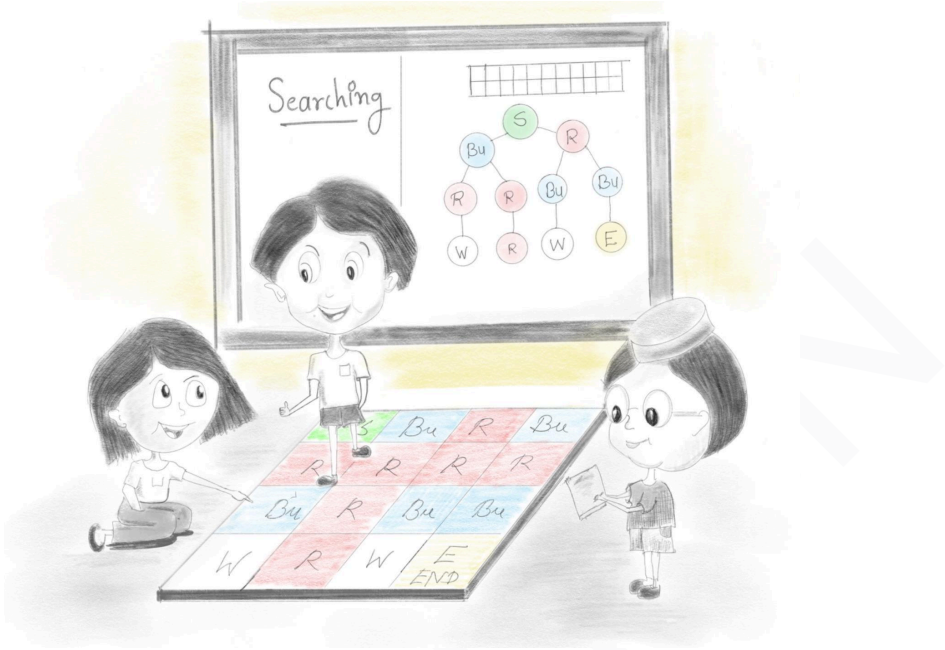
Colour Maze and Searching

Grade 8



A colour maze is a puzzle in which the allowed movement depends on the colour of the square you are currently on. Although the goal is simple—to move from Start (S) to End (E)—the rules create many possible paths.

Some paths reach the goal quickly, some lead to dead ends, and others take longer than necessary. A computer does not solve such puzzles by guessing. Instead, it follows a clear set of rules to systematically explore possible paths. This process is called searching.

To search efficiently, computers first simplify the problem by keeping only the information that matters: the possible positions and the legal moves between them. In this lesson, we will solve a colour maze, represent it as a diagram, and explore different search strategies used in computational thinking.

Note for Teachers: Learning Goals

- Exploration : Apply movement rules to solve a colour maze and identify legal and illegal moves based on square colour and direction.
- Representation: Represent colour maze as a diagram using squares and lines as connections between squares and explain how diagram simply the maze solving.
- Searching: Explore Search strategy to find the path

This lesson can be done in two to three session of 45-minute class or split across four shorter sessions.

- Opening conversation, Introduce the puzzle and Try out puzzle: 30 minutes
- Introduction of the Representation and Searching : 50 minutes
- Discussion, Conclusion and Reflection: 10 minutes

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Colour Maze

Reach from Start (S) to End (E). The colour of a square tells what kind of move is allowed from that square.

Use the following colour movement rules:

W W: White — move straight only - continuing in the same direction as the previous move

Bu B: Blue — move straight OR turn left - continuing in the same direction as the previous move or "Turn Left" depending on the direction in which the player is currently facing.

R R: Red — move straight OR turn right - continuing in the same direction as the previous move or "Turn Right" depending on the direction in which the player is currently facing.

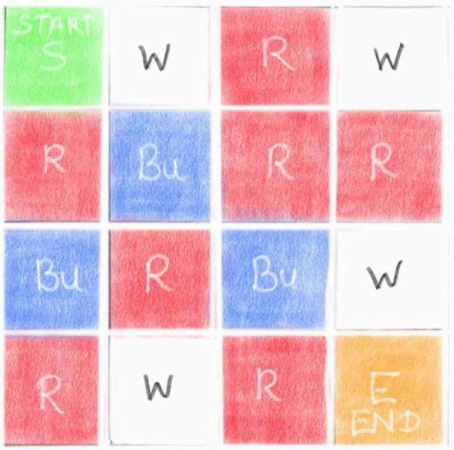


Figure: Reach the End

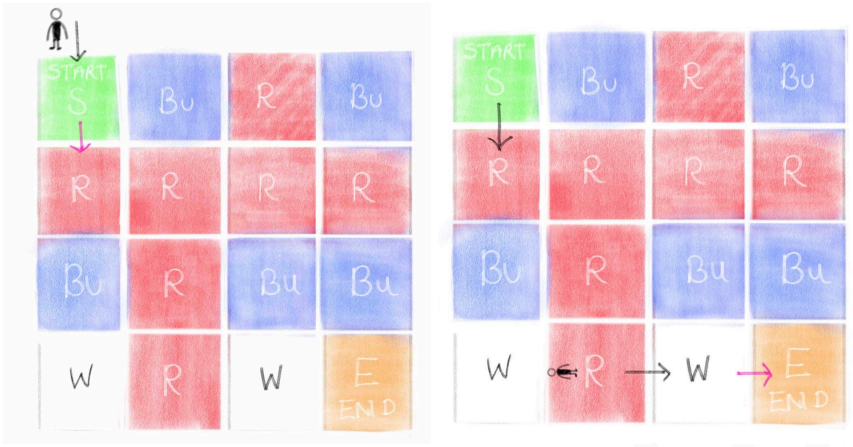
In this puzzle, " Move Straight" means continuing in the same direction as the previous move. "Turn Left" and "Turn Right" depend on the direction in which the player is currently facing. Because of this, solving the maze is not only about position; direction matters as well. And also, one can only move or take left/right turns within the grid.

The same square may behave differently depending on the direction from which it is entered. This makes the puzzle richer than an ordinary maze. If one is moving to the right and enters a white square, the next move must continue to the right. If the player enters a blue square, the next move can either continue straight or turn left. If the player enters a red square, the next move can either continue straight or turn right. This makes the puzzle richer than an ordinary maze. The same square may behave differently depending on the direction from which it is entered.

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Example movement is shown below :
For the START square

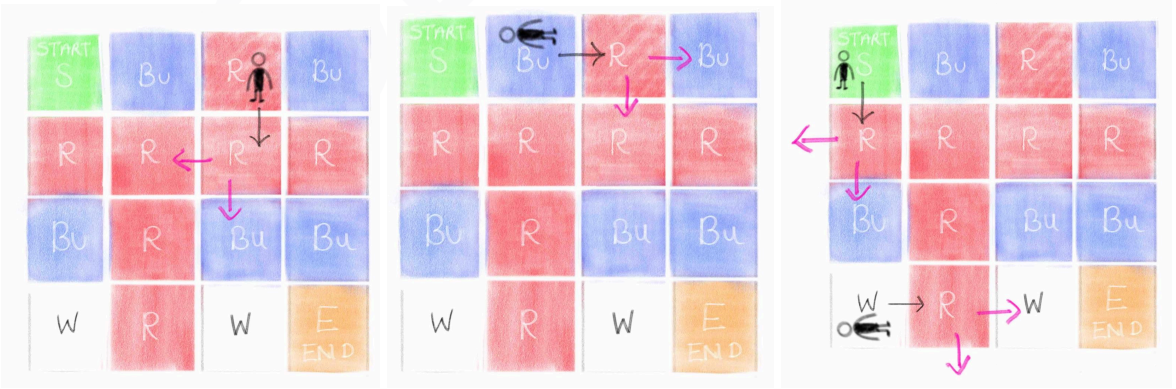
White coloured Square



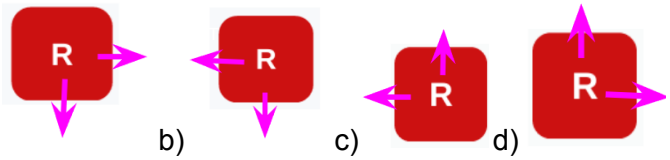
For the blue-coloured square



From the red-coloured square



Q. One enters a red square while moving towards the right. Which moves are allowed?



a) b) c) d)

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Note for Teachers: Classroom Set-Up and Understanding Direction

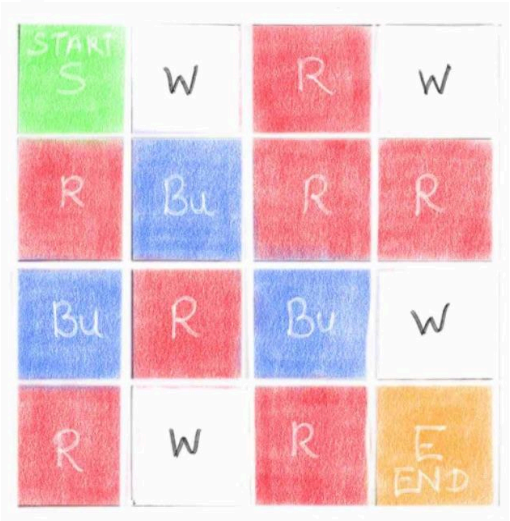
- Draw the colour maze shown as an example on the board or draw it on a floor using chalk. The square tiles are great as they can serve as a grid.
- Explain the puzzle and its rules with an demonstration of given example.
- Begin with a simple activity before Teacher demonstrate the solutions and students starts to solve the other example maze.
  - Ask one student to stand facing forward.
  - Call out colour names one by one.
  - If the colour is white, the student takes one step straight.
  - If the colour is blue, the student may step straight or turn left.
  - If the colour is red, the student may step straight or turn right.
  - Emphasise that turing left and right depend on the direction in which the student is facing.

This activity helps students connect colour instructions with body movement. It also prepares them to understand why direction matters in the maze.

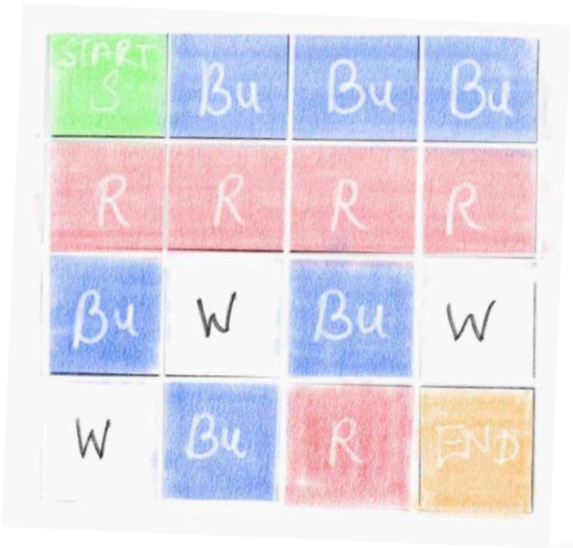
- Invite students to try out the solutions over the grid.
- Once the puzzle is solved, More puzzle can be given for pen-paper exploration.
- Let students experience the puzzle first and get them familiarize the with the direction and let them observe that the same square may behave differently depending on the direction from which it is entered.
- After this section, one can transition into the searching the solution.

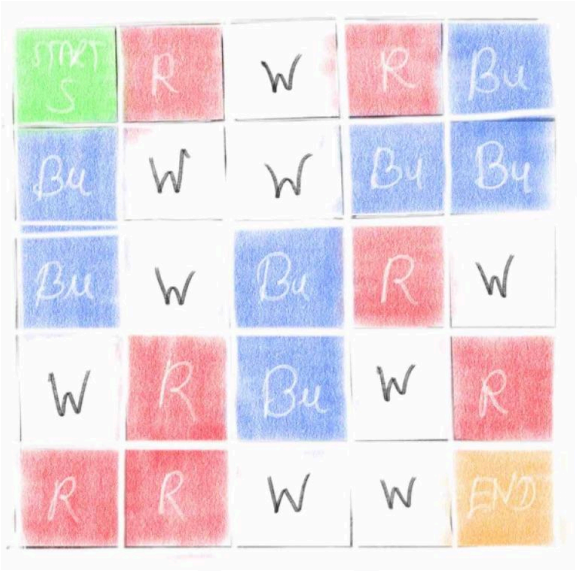
Exploration

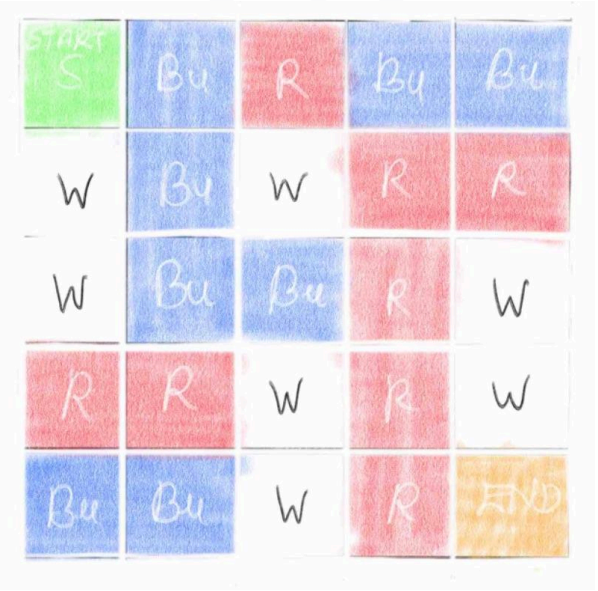
Colour Maze exercises to try in the classroom.



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Note for Teachers: Intro to Searching

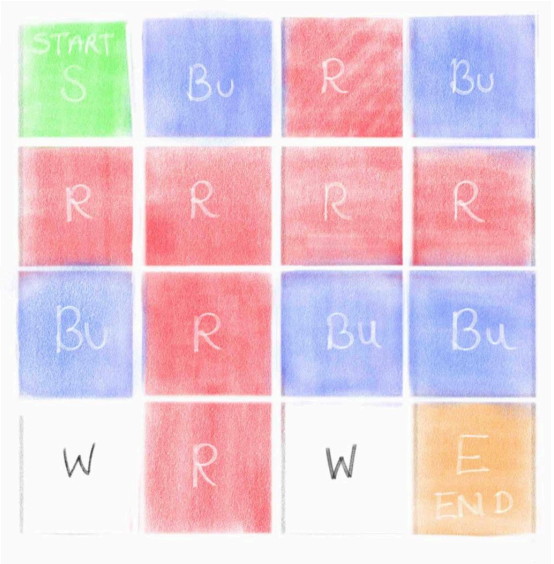
- With this section we would
- Notice that the exact grid drawing style is not important. The important parts are the squares and the legal moves between them, which is the essence of finding the solution of the maze.
- Humans often look at a puzzle and make guesses based on what seems promising. A computer does not do that in the same way. It follows a fixed method that can be repeated every time, even for a very large maze.

Colour Maze: Searching

When solving a maze, we rely on memory and visual clues. Computers, however, need a simpler representation. Instead of focusing on colours and grid lines, one needs to keep only the important information, the possible squares one can go to and the legal moves between squares.

A computer does not "find" the solution to a colour maze the way humans do. Instead, it represents the maze as a diagram with squares and lines. Each square remains a square, and each possible jump or number of steps becomes a connection (line) between the squares. One can solve the maze by simplifying the problem into an easier representation.

Let us take the initial chess maze as an example to understand the searching.

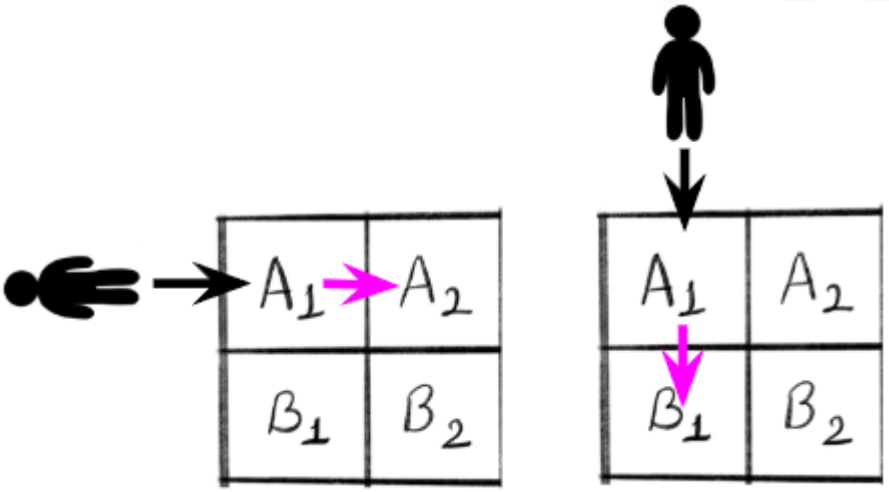


Label the squares in a maze with names such as A1, A2, A3, ... for the first row, B1, B2, B3,... for the second row, and so on. Now we should draw one line for each square and connect each square when a legal move is possible between them.

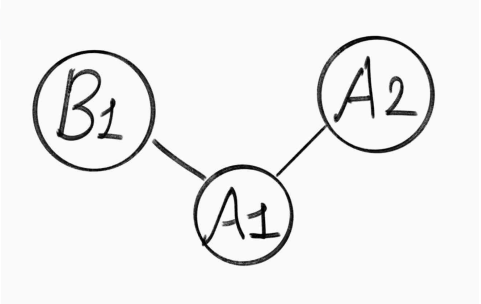
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A_1	A_2	A_3	A_4
B_1	B_2	B_3	B_4
C_1	C_2	C_3	C_4
D_1	D_2	D_3	D_4

Now we need to connect the squares. First, we will take our starting square, A1. A1 should be connected to A2 and B1, since both are one step away vertically/horizontally from A1, depending on the starting direction. We can represent it as A1 → A2, B1.



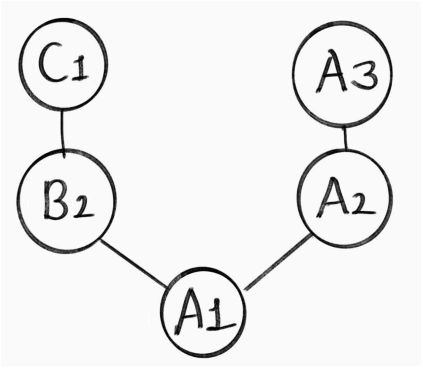
The exact appearance of the diagram is not important. What matters are the squares and the connections between them. We will start to draw the diagram from the start square 'A1'and model the first few branches. Representation of A1 → A2, B1 is shown below.



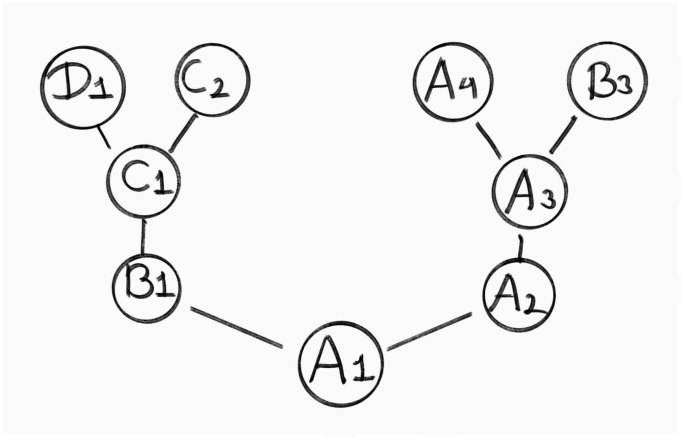
We will do this exercise only for one connection-away square. For example, since we have connected A1 with all connected squares, B1 and A2, we will do the same for B1 and A2.

We can do this exercise for every square. Now, for Square A2 and B1, we will branch out to their connections: A2 → A3 and B1 → C1.

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And, similarly with C1 and A3.

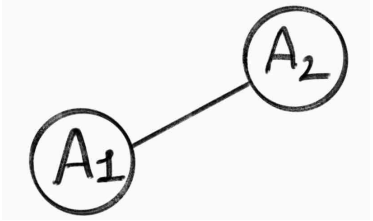


Notice that we are branching till the same breadth at every connection. We will call this method **Breadth-First-Branching**.

Continue drawing the diagram onwards here. All branching will stop once we reach the end square 'D4'.

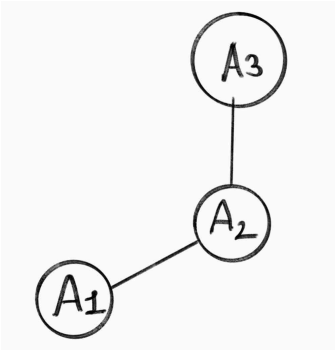
Another way to draw the maze diagram is to make one connection at a time, rather than focusing on every connection from each square and making them all at once. One can focus only on one connection at a time and complete its path.

For example, instead of drawing A1 as A1 → A2, B1, we will only draw A1 → A2 first.

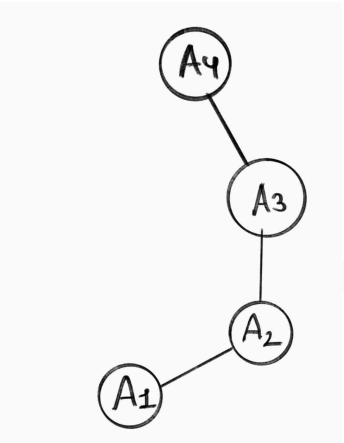


And then we will draw onward A2, such that we will not look for all possible branching, but just one. A2 → A3

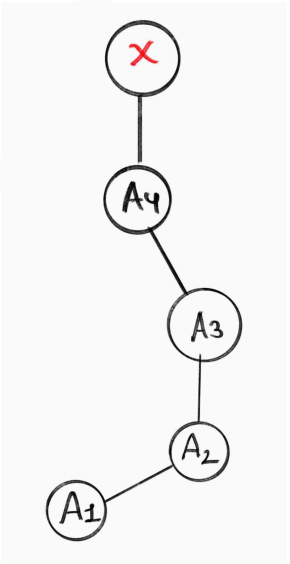
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For A3 → A4



There is no connection onwards A4; in this case, we reach a dead end.

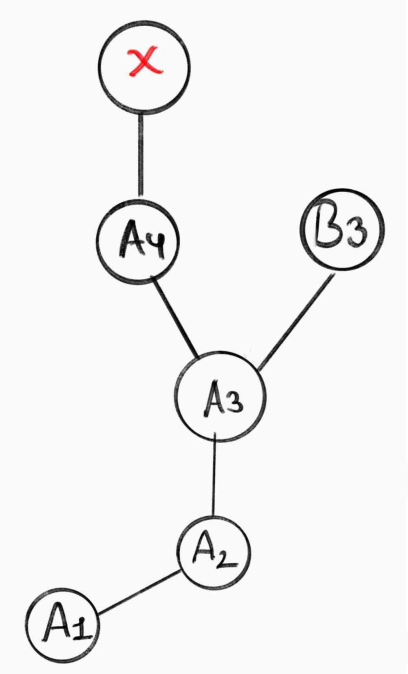


When we hit a dead end, we go one connection back and check if there is another connection from that square.

If there is no other connection with that previous square, we go one step square back further.

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Here, you will see that there is no other connection with A4, so we backtrack to A3. A3 is connected to B3. So we go further with B3.



This is known as **Depth-First-Branching**. Choose one path and keep drawing the connection forward. Ignore all the side branching. Keep going until we find the goal or until one hits a dead end. When you hit a dead end, turn around and walk back to the last branch you ignored.

Now, explore that branch all the way to its end. Do this until you find a solution.

Continue drawing the diagram onwards here. All branching will stop once we reach the end square 'D4'.

Using this type of diagram, not only can we visually see the maze's path, but we can also find the shortest path among all the paths. Once the maze is represented as a diagram using Breadth-first or Depth-first branching, we can use a similar strategy while searching to find the path from Start to End.

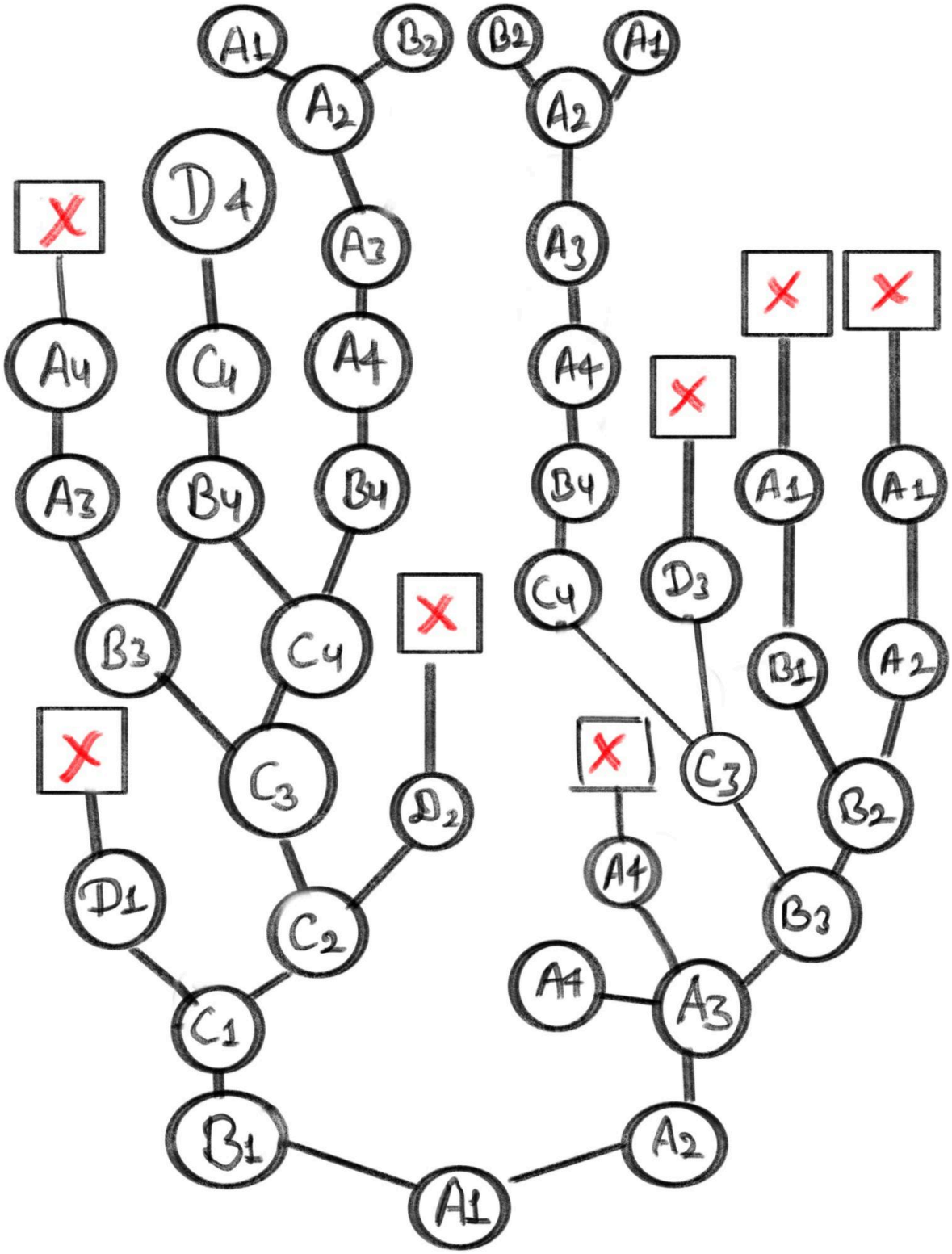
A search algorithm explores:

- Which square to visit next,
- which paths are already explored,
- and which paths lead to dead ends.

Searching means checking possible paths in an organised way until the goal is found. In a maze, there may be several possible moves from one point. A search method gives a clear rule for deciding which move to check first and what to do next.

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Q. For the given colour maze, which method, Breadth-First or Depth-First, finds the solution in the minimum number of steps?



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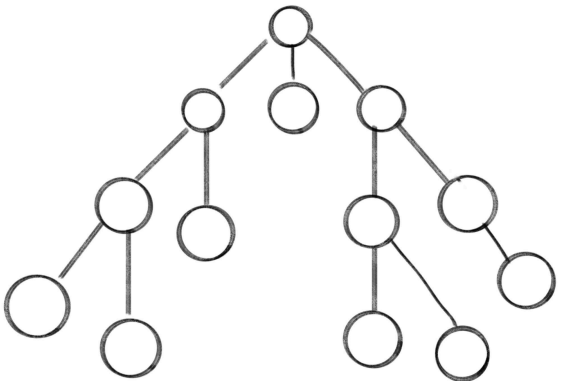
Note for Teachers : Searching Strategy

- Students may focus on making neat drawings. Remind them that the purpose of the diagram is to simplify the maze.
- Encourage students to check whether every connection represents a legal move.
- Searching is a central pillar of computational thinking and helps learners manage complexity by focusing on the information that matters for solving a problem. In this activity, the full grid drawing is not the only thing one need. What matters more is the set of possible positions and the allowed movements between them.
- A diagram is a structure of a hierarchical arrangement of with a squares and lines, where square represent objects or positions and lines represent connections between them.
- One do not need formal graph theory vocabulary at all. The lesson should start with the puzzle experience, then move to naming the simplified representation as a diagram and then they explore the searching strategy to find the solution.

One may compare both the searching methods as follows:

Feature	Breadth-First Search	Depth-First Search
Exploration	Level by level	One path at a time
Backtracking	No	Yes
Finds the shortest path	Yes	Not always
May reach dead ends	No	Yes
Best suited for	Finding shortest paths	Exploring possibilities

Q. Fill out the following diagram by labelling each square 1 through 12 according to the order that breadth-first search would visit the nodes in, 1 as starting from the bottom.

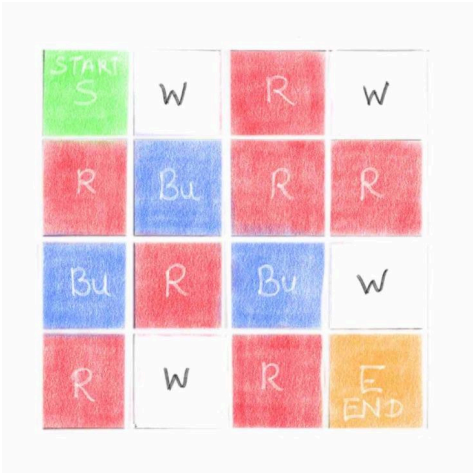


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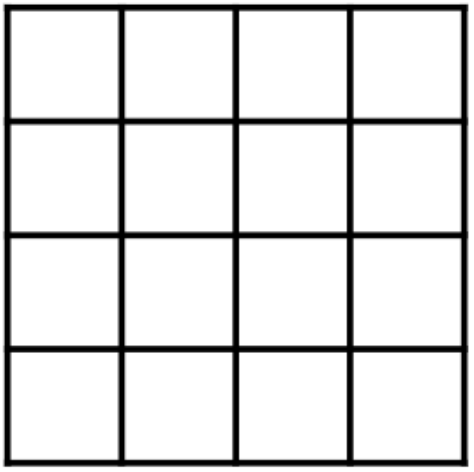
Exploration

Colour Maze exercises to try in the classroom.

A)



Label the maze square

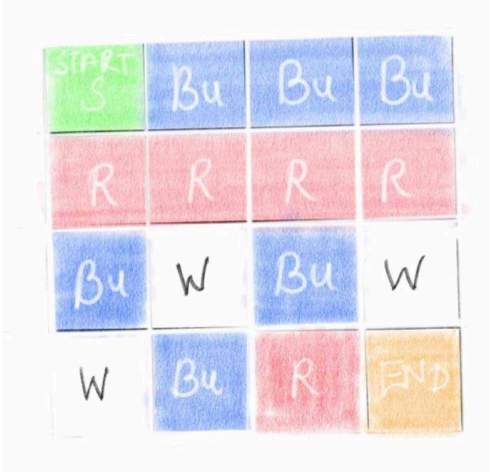


Draw the diagram of the maze with the breadth-first search method

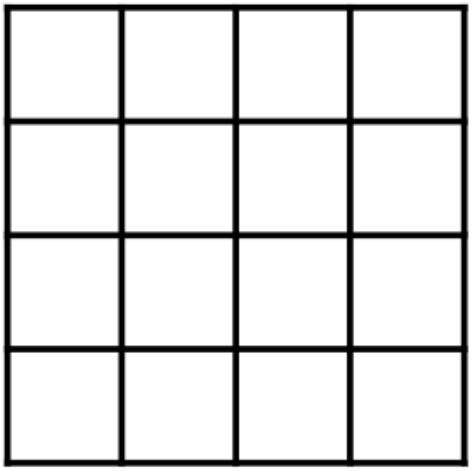
Solution path

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B)



Label the maze square

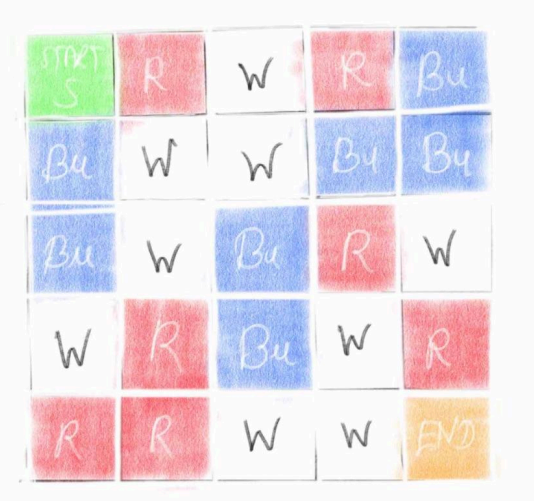


Draw the diagram of the maze with the depth-first search method

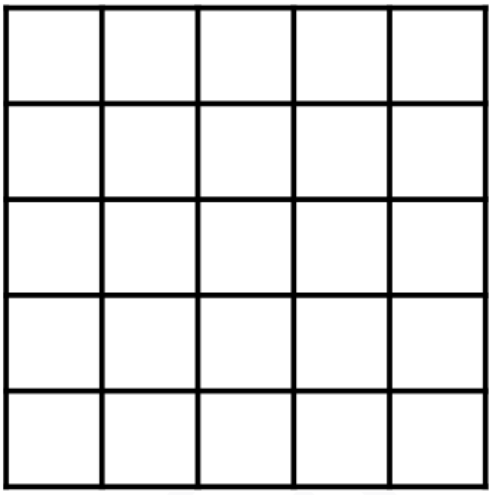
Solution path

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C)

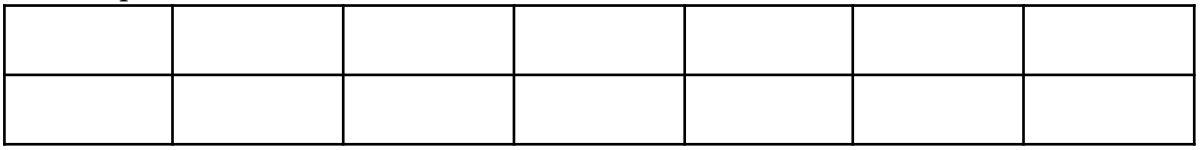


Label the maze square



Draw the diagram of the maze with the depth-first search method

Solution path

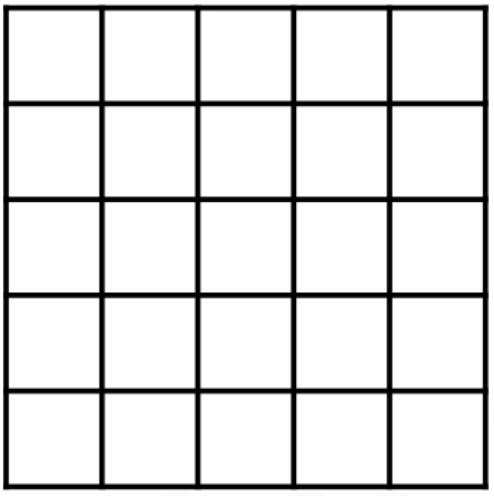


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D)



Label the maze square



Draw the diagram of the maze with the breadth-first search method

Solution path

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Solutions

A.

B.

C.

D.

Discussion, Conclusion and Reflection

Using a colour maze, one can explore how a problem can be solved by following clear rules step by step. The objective was to reach the Goal by moving through a grid from Start to Goal, but the allowed movement depends on the colour of each square. This makes the puzzle an engaging way to introduce the idea of searching in computational thinking.

A path puzzle may look simple at first, but it can contain many possible routes. Some routes reach the goal quickly, some lead to dead ends, and some take longer than necessary. A computer does not guess its way through such a puzzle. Instead, it uses a search method to systematically explore possible moves until it finds a path to the goal.

Searching is a strategy for exploring possible solutions in an organised way. Different search methods are useful in different situations. Breadth-First Search is useful when finding the shortest path is important. Depth-First Search is useful when exploring all possibilities or when computer memory is limited.

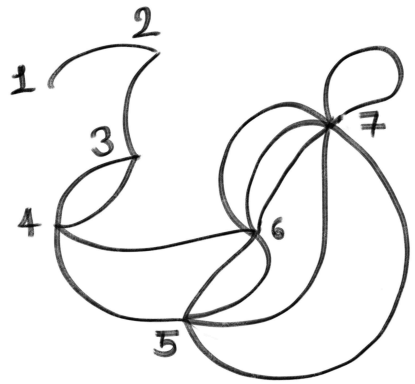
An ant searches for food using a Depth-First Search (DFS) pattern; ants commit to following a single, long trail as deeply as possible. If they hit a dead end, they backtrack to the last intersection and try another path until they find food.

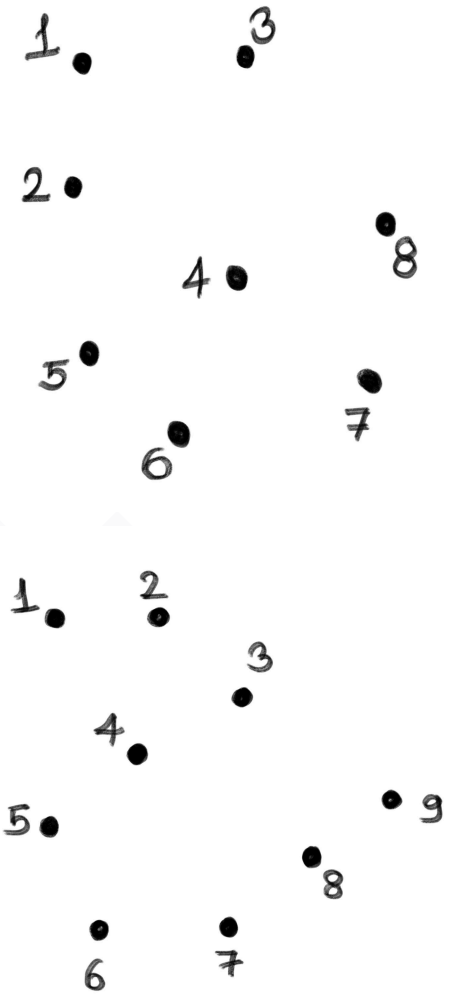
And in the game of chess can virtually go to many moves, the computer only needs to look for possible 8 moves ahead, similar to Breadth-Search (DFS) pattern

By representing a maze as a diagram, we simplify a complex problem and make it easier for computers to solve.

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See and Solve Puzzle





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COMPUTATIONAL THINKING OVER PUZZLE(S)

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